SLEEVEDRIFT · USER REQUIREMENTS

SleeveDrift user requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-02	Amish Chadha	First issue at the kit 1.9.1 upgrade; the fifteen needs are drawn from the users and stakeholders in SVD-PRB-001 and the twelve requirements Amish already holds (R1 to R12); to review with a co-design partner

SleeveDrift user requirements

What the people who would use or stage the kit need, written from their side. They are drawn from the users and stakeholders in SVD-PRB-001 (Tables 1 and 2) and from the Silkyara reports cited there; they were used to score the concept options (SVD-OPT-001). The product requirements that meet them are in SVD-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix SVD-RTM-001 (`03b-traceability.md`). Nothing here is agreed with a co-design partner yet.

> **Safety:** SleeveDrift is an open engineering reference for equipment used by people in a rescue pipe under collapse debris. It is not certified rescue, mining or confined-space equipment.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Safety	Head protection at the face	The person digging at the face has solid cover over the head and shoulders so that a fall of debris does not reach them	High	Hand-mining rescue crews (PRB Table 1)
UR-002	Performance	Spoil leaves the pipe	Spoil is carried out of the pipe at least as fast as the face produces it, so the dig is never waiting for the haul	High	Hand-mining rescue crews; portal crews
UR-003	Usability	The haul is not heavy work	The portal crew can haul with ordinary effort, for a whole shift, without special strength	Medium	Portal crews
UR-004	Safety	Fresh air at the face	Clean air reaches the digger all the time, in every part of the pipe	High	Hand-mining rescue crews; rescue commanders
UR-005	Safety	Bad air is detected	The crew is warned at once of low oxygen, carbon monoxide, hydrogen sulphide or flammable gas	High	Rescue commanders; confined-space practice
UR-006	Performance	A casualty comes out fast	A person who is injured or trapped at the face can be brought out of the pipe within minutes	High	Hand-mining rescue crews; rescue commanders
UR-007	Deployment and assembly	Fits the pipe that is there	Every piece goes in through the pipe and works inside it, with room left for a person to crawl past	High	Hand-mining rescue crews; tunnel contractors
UR-008	Deployment and assembly	Ready within hours	The kit is installed and working within a couple of hours of arriving at the portal	Medium	Rescue commanders
UR-009	Safety	Nothing powered in the pipe	No mains or engine-driven machine inside the pipe, where gas, dust and water collect and a fault cannot be reached	High	Rescue commanders; confined-space practice
UR-010	Environment and durability	Lasts a rescue	The kit keeps working through days of continuous shifts in dust, grit and damp	Medium	Rescue commanders; portal crews
UR-011	Logistics	Moves by road and by hand	The kit travels in cases one or two people can lift, and gets to the portal on foot if the road ends	Medium	Rescue commanders; tunnel contractors
UR-012	Safety	A mistake cannot break the kit on a person	Nobody can overload the ropes and fittings by heaving too hard, and the kit stops before it hurts someone	High	Hand-mining rescue crews; portal crews
UR-013	Cost	Affordable to hold in store	A contractor or a force can afford to hold a kit on standby, as part of emergency planning	Medium	Tunnel contractors; rescue commanders
UR-014	Logistics		A general steel workshop can make the made parts from stock section and	Medium	

ID	Category	Title	Need	Priority	Source
		Buildable and mendable by a workshop	plate, and everything else can be bought		Tunnel contractors; local fabrication workshops
UR-015	Usability	Face and portal can talk	The digger and the portal crew can signal each other at any moment, without batteries or radio	High	Hand-mining rescue crews; portal crews

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.